Guessing π with Random Darts

Work through this on your own. You will find the value of π using nothing but random numbers and Python. About 45 minutes.

Name: _____ Class: _____

You need: Spyder (or any Python editor). Type your code into a new file and press Run. You only need `import random`. Work through the Parts in order and tick them off at the end.

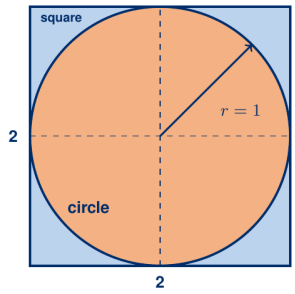
Part 1 The question

1 How could you find out the value of π **without** already knowing it? Write down one idea.

Today you will do it with a dartboard, random numbers and Python. $\pi \approx 3.14159\ldots$ and its digits never end, so any method only gives an **estimate**.

Part 2 What is π? The maths first

π is in every circle's area: **area =** $\pi \times$ **r²**. Here is a circle with radius 1 inside a square that is 2 wide and 2 tall.



2 Area of the circle (r = 1): $\pi \times 1^2$ = _____ Area of the square: 2×2 = _____

3 Imagine throwing darts at random so they land evenly over the square. The fraction that land inside the circle is circle area ÷ square area = _____ ÷ _____

4 If $\pi \div 4 \approx 0.785$, about how many out of 1000 darts land inside the circle? _____

The four quarters of the picture are identical, so we only need **one**. Our dartboard is the top-right quarter: a square 1 wide and 1 tall, with a quarter circle of radius 1 whose centre is the bottom-left corner.

5 Area of the 1 by 1 square: _____ Area of the quarter circle: $\frac{1}{4} \times \pi \times 1^2$ = _____. So the fraction inside is still $\pi \div 4$.

Part 3 Random numbers in Python

Type these two lines into the console (bottom right in Spyder), one at a time:

```
import random
random.random()
```

6 Run `random.random()` three times. Write the numbers you get: _____ _____ _____

7 Every number you got is between _____ and _____.

`random.random()` is **uniform**: every value from 0 to 1 is equally likely. So if we use one random number for x and another for y, the dart lands evenly anywhere in the square.

8 Why would our π come out wrong if the darts bunched up in one corner? Write one sentence.

Part 4 Throw one dart

Type this into a new file and run it:

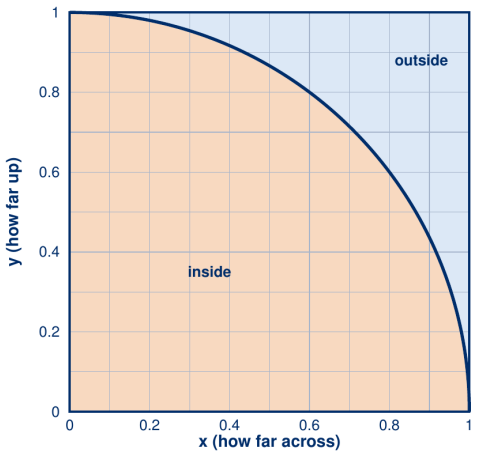
```
import random

x = random.random()   # how far across
y = random.random()   # how far up

print(x, y)
```

9 Run it three times. Write each x and y in the table, then mark each dart with a cross on the dartboard.

Dart	x	y
1		
2		
3		



Part 5 Is the dart inside?

A dart is **inside** if its distance from the bottom-left corner is at most r = 1. By Pythagoras, the distance is $\sqrt{x^2 + y^2}$. Fill in the steps:

Step	Maths
inside means	distance $\leq$ r
square both sides	$x^2 + y^2 \leq$ _____
r = 1, and 1^2 = _____	$x^2 + y^2 \leq$ _____
in Python	`x*x + y*y <=` _____

In Python, `x^2` does **not** square x. Write `x*x` (or `x**2`).

10 Work out each dart by hand. Dart 1 is done for you. Use the rule: inside if x*x + y*y is 1 or less.

Dart	x	y	x*x + y*y	Inside or outside?
1	0.6	0.5	0.61	inside
2	0.9	0.8		
3	0.3	0.7		
4	0.2	0.95		
5	0.95	0.4		

Part 6 Your turn: inside or outside in code

Fill in the two blanks, then type it in and run it **five** times.

```
import random

x = random.random()
y = random.random()

if _____________________________:
    print("inside!")
else:
    print("_____________")
```

11 Out of your 5 runs, how many said inside? _____ About how many would you expect? _____

Hints (upside down on purpose: try Hint 1 first)

Stretch: use a `for` loop to throw 10 darts. Hint 1: the first line is `for i in range(10):`. Hint 2: everything you want repeated is indented underneath it.

Part 7 Count 10,000 darts

Now throw lots of darts and **count** the ones that land inside. `inside = inside + 1` adds one to the count. Fill in the three blanks and run it three times.

```
import random

total = _____________
inside = _______

for i in range(_________):
    x = random.random()
    y = random.random()
    if x*x + y*y <= 1:
        inside = ___________________

print(inside)
```

12 Write the three numbers it printed: _____ _____ _____

13 Divide your first number by 10,000: _____. Now use a calculator for $\pi \div 4$ (use 3.14159): _____. Are they close?

Part 8 The trick

We know that **inside** ÷ **total** ≈ $\pi \div$ **4**. Multiply both sides by 4:

14 $\pi \approx$ _____

Add these two lines at the **end** of your program (not indented):

```
pi_guess = 4 * inside / total
print(pi_guess)
```

15 Run it three times and fill in the table. For the last column, work out how far each guess is from 3.14159.

Run	My guess for π	How far from 3.14159?
1		
2		
3		

Part 9 More darts, better guess

16 Change **only** the number in `total = ...` and fill in the table. One million darts takes a few seconds.

total	Guess 1	Guess 2	Guess 3
100			
10,000			
1,000,000			

17 What do you notice as the number of darts gets bigger? Write one or two sentences.

18 Why can this method never give π exactly?

Part 10 This has a name

A **Monte Carlo method** uses lots of random tries to answer a question. It is named after a famous casino in Monaco. People use it for weather forecasts, video game AI and physics simulations.

19 Think of something else that is too hard to work out exactly, but that you could estimate by trying it thousands of times at random. Write it down.

Hints (upside down on purpose: try Hint 1 first)

Stretch: print the guess every 1,000 darts so you can watch it settle. Hint 1: do this inside the loop. Hint 2: `i % 1000 == 0` is true on every 1,000th dart. Hint 3: `if i % 1000 == 0:` then, indented, `print(4 * inside / (i + 1))`.

Finished What I can do now

- ☐ I know `random.random()` is uniform, so darts land evenly
- ☐ I know the distance from the corner is $\sqrt{x^2 + y^2}$
- ☐ I know inside means `x*x + y*y <= 1`
- ☐ I know the quarter circle is $\pi \div 4$ of the square
- ☐ I know $\pi \approx 4 \times$ inside $\div$ total